PROBLEM STATEMENT: PROJECTILE MOTION SIMULATOR GUI

1. Problem Being Addressed

In classical physics education, understanding two-dimensional projectile motion relies heavily on analytical formulas and static textbook diagrams. Students and researchers often struggle to develop an intuitive grasp of how initial conditions specifically initial velocity and launch angle interact dynamically to influence trajectory path, peak height, total flight time and horizontal distance.

Furthermore, manual calculations are repetitive and prone to input errors, while standard computational software often lacks a simple, accessible Graphical User Interface (GUI) designed for quick parameter tweaking and visual feedback.

2. Objective of the Application

The main objective of this application is to provide an interactive, lightweight graphical user interface built entirely within Scilab. The software enables users to seamlessly simulate and visualize ideal projectile motion (assuming no air resistance and standard gravity, $g = 9.81\ m/s^2$).

Key objectives include:

• Instant Parameter Computation: Automatically calculating precise kinematic metrics upon user input.

• Live Trajectory Visualization: Generating a clear 2D trajectory plot complete with dedicated visual markers for the peak height (apex) and landing point.

• Input Validation & Error Handling: Ensuring application stability by catching invalid inputs (such as non-numeric characters or launch angles outside the logical range of $0° < \theta < 90°$).

Theoretical Kinematic Formulas:

• Time of Flight (T) = $(2 * v * \sin(\theta)) / g$

• Maximum Height (H) = $(v^2 * \sin^2(\theta)) / (2 * g)$

• Horizontal Range (R) = $(v^2 * \sin(2\theta)) / g$

3. Intended Users and Use Cases

• Physics Students & Learners: Serves as an interactive educational aid to explore "what-if" scenarios, visualize parabolic curves and verify manual physics coursework.

• Educators & Instructors: Acts as a visual demonstration tool during lectures or laboratory sessions to show real-time changes in projectile motion.

• Scilab Developers: Provides an exemplary model for building modern, responsive GUIs and dynamic plotting routines within the Scilab programming environment.